Camera Feed on Background — Unity Asset by makaka.org

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Camera Feed on Background — Unity Asset that uses Back/Rear or Front/Face Camera on the player's mobile device to display the Live Video Input on the Background of the Scene. **Safety First.**

It doesn't use any Motion Sensor of a Mobile Device (Gyroscope or Accelerometer) to rotate the Game World like in AR Camera Lite (docs) or in AR + VR: MR Camera (docs).

Note: The asset has a free external dependency.
Read the "Getting Started" Tutorial to Import it correctly.

Check out the Support section if you need help.

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Features of Camera Feed

Bring the enchanting Power of Camera Feed into your amazing Game or App:

- ⭐ Cross-platform Camera Feed: iOS, Android.
- ⭐ Device Orientations: Portrait, Landscape.
- ⭐ Smartphone Cameras: Rear/Back, Front/Face/Selfie.
- ⭐ Button: Switch Camera (Back/Front).
- ⭐ Instant Launch after granting the camera permission.
- ⭐ Quick Testing in Unity Editor with Webcam.

Package Contains

- ⭐ Demo Scene with Camera Feed.
- ⭐ Menu Scene with Safety Tutorial.
- ⭐ Permission Scene with Camera Permission Request using free Native Camera.
- ⭐ Loading Screen to switch scenes seamlessly.

Check the Map of Unity Assets to choose the product that best suits your needs.

Limitations

Asset doesn't operate with Native Code and relies on WebCamTexture class.
Any possible Limitations are related to this class.

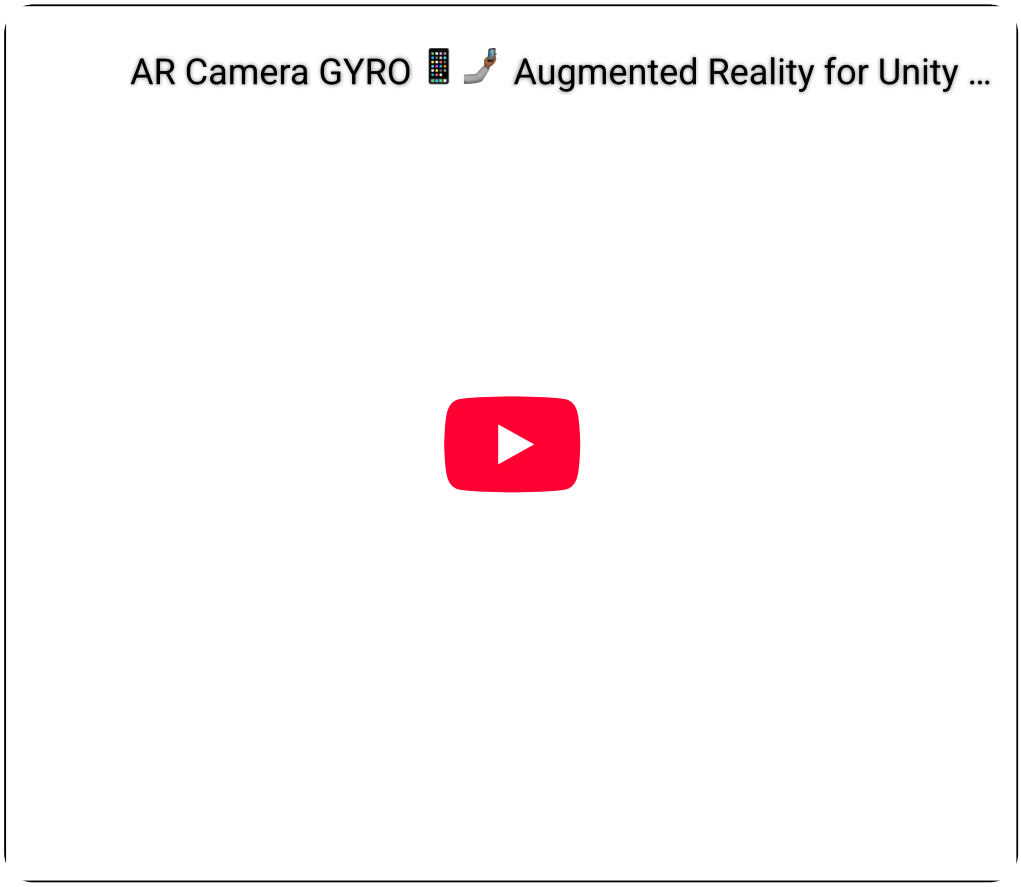
Known Issues

- ⭐ *Black Screen on iOS*. ***Solution:*** You just need to fill Camera Usage Description in Unity Editor:
  File > Build Profiles > Player Settings > iOS > Other Settings > Configuration> Camera Usage Description.

Use Cases

1. With adding rotation of the Game Camera with the Motion Sensor of a Mobile Device (Gyroscope or Accelerometer), Camera Feed on Background can be used as a “Pseudo AR Camera” to display 2D or 3D objects as though they were in the real world.

This technique was implemented in AR Camera Lite (docs).



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2. With integration of Google Cardboard XR Plugin, the Camera Feed on Background can be used as a Mixed Reality (MR) Camera. This idea was implemented in AR + VR: Mixed Reality Camera (docs).

Unity AR + VR ⭐ Mixed Reality (MR) with Google Ca...

3. Previous version of AR FPS Shooter (docs) used Pseudo AR mechanics that you can watch below.

AR Shooter — Unity Asset 🎯 Augmented Reality for ...

Tutorial

> *This tutorial is relevant for Camera Feed on Background 3.0.*
> *Tutorial for the previous version can be found only in the asset folder.*

Getting Started with Camera Feed on the Background

Folders & Files in the package by default:

- ⭐ Makaka Games.

Steps

> *If you have any issues with the first launch then just Reach Support with Invoice Number and Get Help.*

1. Create a New Unity Project with Unity 6000.3.7 & “Universal 3D” Template.

2. File > Build Profiles > Platforms > Android, iOS > Switch Platform.

3. Next Packages must be manually installed with Git & Unity Package Manager. **I recommend using indicated versions to get a guaranteed experience**:

   - ⭐ Native Camera 1.5.0.
     Hash value for installing: 72f5ebba26c219b4c4cda202bff2f59bfdaa47f8

4. Download and import the Camera Feed on Background into Unity.

5. Window > TextMeshPro > Import TMP Essential Resources.

6. Reopen Unity Project.

7. Open Scene: Makaka Games > Camera > CameraFeedOnBackground > Scenes > Demo.

8. File > Build Profiles > Scene List > Add Scenes:

   - ⭐ *Menu*,
   - ⭐ *Permission_Camera* (from Makaka Games > Publisher > Permissions > Camera > Scenes),
   - ⭐ *LoadScreen* (from Makaka Games > Publisher > SceneControl > Scenes),
   - ⭐ *Demo*.

9. ***Option:*** Limit the Screen Orientation from Auto (by default) to Portrait only or Landscape only by indication of appropriate Scene Loading Methods on the corresponding buttons that cause a transition to another scenes. Orientation in the Player Settings is used for the 1st scene in the build.

10 ***Option:*** File > Build Profiles > Development Build > Check (to See the Log in the Console).

11 File > Build Profiles > Player Settings > iOS > Other Settings > Camera Usage Description > Fill it (any notice for your app).

12 Test in the Unity Editor or Build.

Useful Article: How to Test iOS App without Developer Account?

Placing Visible Game Objects

If you need to place visible Non-UI Game Objects over the Device Camera Feed, then just Activate the Game Object called *Game World*, and place Game Objects inside it.

Game World contains a separate Camera for this task: *Main Camera*.

After activating, you will have 2 Game Cameras in the Scene:

- ⭐ *Background Camera* (*UI* Layer) for Device Camera Feed and UI.
- ⭐ *Main Camera* (other Layers) for visible Non-UI Game Objects.

> *Game Objects will not move based on Device Movement.*
> *Use AR Camera Lite (docs) or AR + VR: MR Camera (docs) instead.*

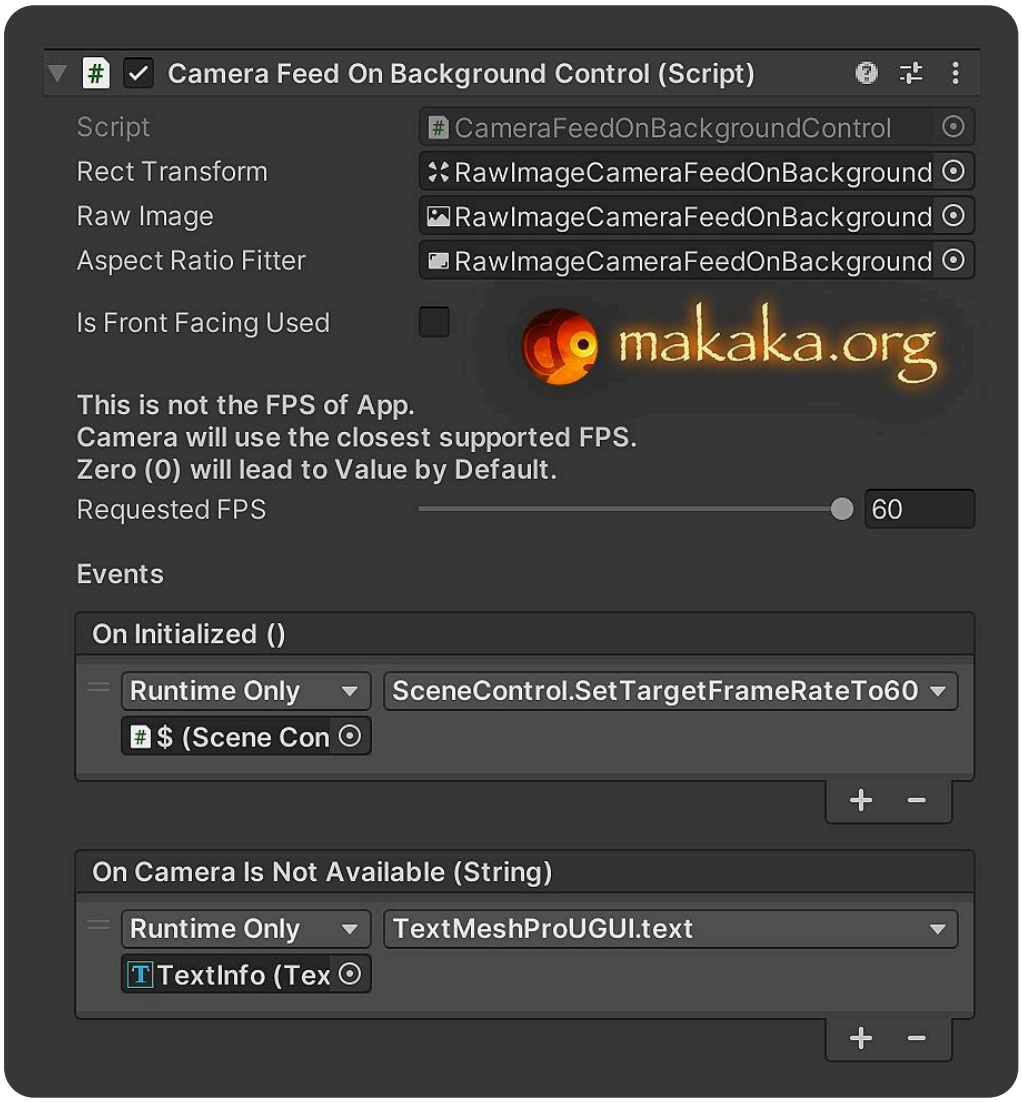
Feed FPS and App FPS

The Unity Package has a Prefab called *CameraFeedOnBackground* that is already customized for the best experience in the *Demo* Scene.

You can customize the Prefab Instance on the Scene with Target FPS of Camera Feed (1) and App (2) —
these 2 types of Frames Per Second (Frame Rate) are set independently of each other in the inspector of *CameraFeedOnBackgroundControl.cs* script:

1. *Requested FPS:* Camera Feed will use the closest supported FPS. Zero (0) FPS will lead to Value by Default.

2. The App will strive to use Target FPS that is set in the **Events** section.



Testing

You can test Camera Feed quickly in the Unity Editor without a Smartphone through physical Web Camera.

Tested with Platforms

Mobile Platforms:

- **iOS** on iPhone 15;
- **Android** on Samsung Galaxy A71.

If you need WebGL support, please use a 3rd party asset called Device Camera WebGL.

Support

First, read the latest docs online.
If it didn't help, get the support:

- Unity Discussions.
- Email.

Changelog

Check the current version of Camera Feed on the Asset Store.
The latest versions will be added as soon as possible.

3.0 (Unity 6.3):

Improvements:

- Unity 6000.3.7.
- URP (Universal Render Pipeline) to improve performance (instead of the Built-in Render Pipeline).
- Input System (New Standard in Unity) for All Project-related Scripts: *instead of Input Manager*.
- Namespaces for All Scripts.

Fixes:

- ⭐ All Bugs related to the Unity Engine itself like WebCamTexture lagging and others were fixed without notices (I didn't observe them during the last tests).

2.0:

Features:

- ⭐ Button: Switch Camera (Back/Front).
  - ⭐ Unity Event *OnCameraSwitched()* is exposed in Inspector.

Improvements:

- ⭐ Unity 2022.3.18.

1.3:

Features:

- ⭐ Front Camera Support.

1.2:

Improvements:

- ⭐ Unity 2021.3.21.

Fixes:

- ⭐ Zoom Effect of Camera Feed in Vertical/Portrait Mode.

1.1:

Features:

- ⭐ Horizontal/Landscape Mode support.